MILITARY SPECIFICATION

CONTROL UNIT, ALARM SET:

C-9412/FSS-9(V)

This interim amendment is approved for use within the US Army Belvoir Research, Development, and Engineering Center, Department of the Army, with MIL-C-0052853C(ME), dated 19 November 1984.

PAGE 2

2.1.1, under SPECIFICATIONS, MILITARY, delete:

"MIL-G-20241 - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure Sensitive.

MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive."

PAGE 3

2.1.2, under DRAWINGS, ME, delete "TA13220E4200" and substitute "TA13220E4085"

2.2, delete and substitute:

"2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained."

3.1, line 2, delete "TA 13220E4200" and substitute "TA 13220E4085".

PAGE 4

3.2, delete and substitute:

"3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the control unit from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish control units conforming to this specification."

AMSC N/A FSC 6350

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3.3.1.1, delete and substitute:

"3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimila metals and methods of protection are defined and detailed in MIL-STD-889."

PAGE 7

3.5.3a, line 3, delete "ms" and substitute "milliseconds (ms)".

3.5.6.4, line 4, delete period after "switch" and insert: "and alarm outputs to the AA shall be inhibited when the CU mode switch is in the test/reset position."

Add new paragraph 3.5.6.5:

"3.5.6.5 AA inhibit. With a low resistance (less than 100 ohms) applied frc terminal 7 on terminal board TB8-F to ground, tamper alarm inputs shall be inhibited from producing alarms at the delayed and AA outputs when the CU mode switch is in the secure or access mode."

PAGE 12

3.5.11.2, in table, output number 4: Change "4 to (TB1-6" to "4 (TB1-6".

PAGE 13

3.5.11.4, line 3, delete "unless specified (see 6.2)".

PAGE 14

3.5.13a, line 3, delete "4 or 7 result" and substitute "4 or 7 shall not result.

3.5.13g, line 2, delete and substitute: " to or less than 18.3 or equal to c greater than 21.7 Vdc, the".

PAGE 15

3.5.14.3, line 4, delete "18 and 20" and substitute "18 to 20".

PAGE 16

3.5.15.1, line 9 delete "impedence conditions" and substitute "impedance or voltage conditions".

Line 13, delete "Impedance" and substitute "Impedance or Voltage".

PAGE 17

3.5.16.3, line 1, delete "18 and 20" and substitute "18 to 20".

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3.10, delete in its entirety and substitute:

"3.10 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantitites indicated will be loaned by the Government (see 6.8) for manufacturing and testing modules and major components when furnished separately:

Item No.	Description	Identification	Quantity
1.	Stable base drawing of PWB, Tamper/Alarm	13220E4169, Sh 2	1 (see note a,b)
2.	Stable base drawing of PWB, Voltage Monitor	13220E4174, Sh 2	1 (see note a,b)
3.	Stable base drawing of PWB, Audible Alarm	13220E4175, Sh 2	1 (see note a,b)
4.	Stable base drawing of PWB, Instantaneous Alarm	13220E4177, Sh 2	1 (see note a,b)
5.	Stable base drawing of PWB, Latched Alarm	13220E4178, Sh 2	1 (see note a,b)
6.	Stable base drawing of PWB, Alarm I.D.	13220E4179, Sh 2	1 (see note a,b)
7.	Stable base drawing of PWB, LED	13220E4180, Sh 2	1 (see note a,b)
8.	Stable base drawing of PWB, Interconnect	13220E4172, Sh 2	1 (see note a,b)
9.	Stable base drawing of PWB, Power Supply	13226E1154, Sh 2	1 (see Note a,b)
10.	Control Unit, Alarm Set	C-9412/FSS-9(V) NSN 6350-00-228-2735	1 (see note c)

NOTES:

a. Provided when procuring Control Unit.
b. Provided when it's associated assembly procured as major component procured separately.
c. Provided when procuring power supply separately and EMI testing of first article is required."

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PAGE 21

Add the following new paragraph:

"4.1.4 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material."

PAGE 33

4.6.1.2, line 3, delete "50071" and substitute "5007.1".

4.6.2.3, line 2, delete "2.5" and substitute "1.0".

PAGE 35

5.3.1.1, line 8 after "I or IV," add "finish B,".

lines 8 and 9, delete last two sentences.

PAGE 36

5.4.1, line 3, delete last sentence.

PAGE 37

6.2d, delete in its entirety.

6.2h, delete "degree" at two places and substitute "level".

6.2i, delete in its entirety.

PAGE 41

3.e, add to list of switches:

"S9 ON (closed) AA tamper"

3., add following step g:

"g. Adjust potentiometers R6 on A9 PWB and R8 on A10 PWB fully clockwise."

3.1, lines 5 and 6, delete "temperature (65° to 82°)" and substitute temperature between 65 °F and 82 °F."

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PAGE 43

4.1.1e, delete in its entirety.

4.1.1f, delete in its entirety.

4.1.1g, line 1, delete first sentence.

PAGE 45

4.1.8, delete in its entirety.

PAGE 46

4.2j, line 3, delete "18.5" and substitute "18.2".

4.2n, line 3, delete "21.5" and substitute "21.8".

4.2u, delete in its entirety and substitute:

"u. Set timer function switch S3 to AC FAIL (position 6), stop trigger switch S5 to position 2, DVM switch S2 to position 2, and monitor switch S1 to position 6, and reset time interval meter. The DVM voltage shall be less than 0.5 Vdc."

4.2v, lines 2-4, delete last sentence and substitute:

"Within 5 seconds the timer shall stop and indicator light AC FAIL shall illuminate. The DVM voltage shall be greater than 7.0 Vdc."

PAGE 47 and 48

4.4, Delete in its entirety and substitute:

"4.4 Secure Mode Test.

a. Set latch/nonlatch switch on status processor to LATCH.
b. Set timer function switch S3 to INST, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
c. At least 12 seconds after performing step e of 4.3, momentarily depress reset switch S8 and reset time interval meter.
d. Momentarily set sensor simulator switch A1 to OPEN to simulate an intrusion alarm. Alarm indicator light INST shall illuminate, the proper LED on PWB A12 shall illuminate, and the time interval meter shall begin running.
e. When the timer stops, the meter shall indicate 10 ±2 seconds. Alarm indicator lights DELAYED and AA shall illuminate within ± 4 seconds of the time at which the timer stops. It is not necessary to accurately measure the ± 4 seconds.
f. At least 12 seconds after alarm indicator lights DELAYED and AA illuminate, momentarily depress S8 alarm indicator light INST shall extinguish.

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NOTE: All alarm indicator lights will be off while S8 is depressed. Throughout these procedures, it is understood that the condition described in a given step immediately afte momentarily depressing S8 is the condition that shall exist after the release of S8.

g. Turn CU mode switch to TEST/RESET, THEN momentarily depress S8. Alarm indicator lights DELAYED and AA shall extinguish.

h. Turn CU mode switch to SECURE and reset time interval meter. LED A1 on PWB A12 shall extinguish.
NOTE: Throughout this and subsequent tests, disregard the status of indicator light A/S, unless otherwise specified.

i. Set S3 to Egress pos. and S5 to pos. 2 and reset the time interval meter if necessary.

j. At least 12 seconds after last turning mode switch to SECURE, momentarily open sensor simulator switch B1. Alarm indicator light INST and the proper LED on the A12 PWB shall illuminate and the time interval meter shall begin running.

k. When the time interval meter stops, alarm indicator lights DELAYED AND AA shall illuminate. The time interval meter shall indicate 10 $\pm$ 2 seconds.

l. As soon as the time interval meter stops, turn the CU mode switch to TEST/RESET and back to SECURE, then momentarily depress S8 and reset time interval meter. Alarm indicator lights DELAYED and AA and LED on PWB A12 shall extinguish. Alarm indicator light INST may extinguish. If it does not, wait 4 seconds and momentarily depress S8. Alarm indicator light INST shall extinguish.

m. Repeat steps j.,k., and l., for sensors simulator switches C1, D1, and E1.

n. Set S3 to INST pos. and S5 to pos. 1 and reset time interval meter.

o. AT least 12 seconds after last returning the mode switch to SECURE, momentarily set sensor simulator switch A3 to OPEN to simulate a tamper alarm. Alarm indicator lights INST, DELAYED, an AA shall illuminate and the time interval meter shall begin running. Non of the LEDs on PWB A12 shall illuminate.

p. When the timer stops, it shall indicate 10 $\pm$ 2 seconds.

q. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish. The DELAYED and AA alarms indicator lights shall be lit when the switch is released.

r. Turn mode switch to TEST/RESET and back to SECURE and reset time interval meter.

s. Momentarily depress reset switch S8. After release of S8, all alarm indictor lights shall be off.

t. Repeat steps o. through s. using sensor simulator switches B3, C3, D3, E3, F3, AA tamper switch S9, dialer switch S10, and CU tamper switch. In one of these trials, allow 4 seconds to elapse after performing step p and before step q.

u. At least 12 seconds after last turning mode switch to SECURE, momentary set sensor simulator switch F1 to OPEN to simulate a duress alarm. Alarm indicator lights INST and DELAYED shall illuminate, the time interval meter shall begin running, and the LED F on PWB A12 shall illuminate. The AA indicator light shall not light.

v. When the timer stops, it shall read 10 $\pm$ 2 seconds.

w. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.

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x. Turn CU mode switch to TEST/RESET and back to SECURE and momentarily depress S8. Alarm indicator light DELAYED AND LED F on PWB A12 shall extinguish. Reset time interval meter.
y. Set latch/nonlatch switch on status processor to NONLATCH.
z. Set timer function switch S3 to LATCH/NL. Reset time interval meter.
aa. At least 12 seconds after last turning mode switch to SECURE, momentarily set sensor simulator switch A1 to open. Alarm indicator light INST shall illuminate and LED A on PWB A12 shall illuminate. After 10 ±2 seconds (it is not necessary to measure this time accurately) alarm indicator lights DELAYED and AA shall illuminate and the time interval meter shall start. When the time interval meter stops, it shall read 10 ±2 seconds.
bb. Momentarily depress reset switch S8. Alarm indicator lights INST and DELAYED shall extinguish.
cc. Turn CU mode switch to TEST/RESET and back to SECURE, momentarily depress S8, and reset time interval meter. Alarm indicator light AA and the proper LED on PWB A12 shall extinguish.
dd. Repeat steps o. through s. Responses shall be as before, except in step q., both alarm indicator lights INST and DELAYED shall extinguish.
ee. Repeat steps u. through x. Responses shall be as before except, in step w., alarm indicator light DELAYED shall also extinguish.
ff. Momentarily set sensor simulators A1 to OPEN, wait 1.5 ± seconds than switch B2 to OPEN. LED A on board A12 shall illuminate and LED B shall not illuminate. Disregard alarm indicator lights.
gg. Turn mode switch to TEST/RESET and back to SECURE. (Allow for exit time before proceeding.)
hh. Momentarily set sensor simulator switch A1 to OPEN. Wait 700 ±100 ms then set switch B1 to OPEN. LED's A and B shall illuminate. Disregard alarm indicator lights.
ii. Push transmitter resync switch on status processor. The test fixture RESYNC indicator light shall illuminate while switch is depressed."

PAGE 49

4.5,a, delete and substitute:

"a. Set latch/nonlatch switch on status processor to LATCH and set CU mode switch to ACCESS position.
b. Set timer function switch S3 to INST, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
c. Momentarily depress reset switch S8 and reset time interval meter.
d. Momentarily set sensor simulator switch A1 to open. Wait at least 12 seconds. None of the LEDs on the test fixture shall illuminate. LED A on PWB A12 shall illuminate.
e. Turn CU mode switch to SECURE and return to ACCESS. LED A on PWB 12 shall extinguish.
f. Momentarily set sensor simulator switch A3 to OPEN. Alarm indicator lights INST, DELAYED, and AA shall illuminate and the time interval meter shall begin running. None of the LED's on PWB A12 shall illuminate.
g. When the timer stops, it shall indicate 10 ±2 seconds.

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h. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.
i. Turn mode switch to TEST/RESET and back to ACCESS. Reset time interval meter.
j. Momentarily depress reset switch S8. Alarm indicator lights DELAYED and AA shall extinguish.
k. Momentarily set sensor simulator switch F1 to OPEN to simulate a duress alarm. Alarm indicator lights INST and DELAYED shall illuminate, the time interval meter shall begin running, and LED F on PWB A12 shall illuminate. The AA indicator light shall not light.
l. When the the timer stops, it shall indicate 10 $\pm$2 seconds.
m. Momentarily depress reset switch S8. Alarm indicator light INST shall extinguish.
n. Turn mode switch to SECURE and back to ACCESS.
o. Momentarily depress reset switch S8, and reset time interval meter. Alarm indicator light DELAYED and LED F shall extinguish.
p. Set latch/nonlatch switch on status processor to NONLATCH.
q. Set timer function switch S3 to LATCH/NL.
r. Repeat steps f through j. Responses shall be as before, except, in step h., when S8 is pushed, alarm indicator lights INST and DELAYED will extinguish and AA will remain on after S8 is released.
s. Repeat steps k. thru o. Responses shall be as before, except, in step m., alarm indicator lights DELAYED and INST shall extinguish."

4.5b through 4.5.s: renumber original steps as steps "t. through kk."

4.6., Delete in its entirety and substitute:

"4.6 Test/reset mode test.

a. Set latch/nonlatch switch on status processor to LATCH and set mode switch to TEST/RESET POSITION.
b. Set timer function switch S3 to INST, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
c. Momentarily depress reset switch S8 and reset time interval meter.
d. Momentarily set sensor simulator switch A1 to OPEN to simulate an intrusion alarm. Alarm indicator lights INST and DELAYED shall illuminate, LED A on board A12 shall illuminate, the sonalert shall begin sounding, and the time interval meter shall begin running. The AA indicator light shall not illuminate.
e. When the timer stops, the sonalert shall stop sounding and the meter shall indicate 10 $\pm$2 seconds.
f. Momentarily depress reset switch S8. All alarms indicator lights shall extinguish. LED A shall not extinguish.
g. Turn mode switch to SECURE and back to TEST/RESET and reset time interval meter. LED A shall extinguish.
h. Momentarily set sensor simulator switch A3 to OPEN (tamper alarm). Alarm indicator lights INST and DELAYED shall illuminate, the sonalert shall begin sounding, and the time interval meter shall begin running. The AA indicator light and LED A on PWB A12 shall not illuminate.

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i. When the timer stops, it shall indicate 10 ±2 seconds. The sonalert shall stop sounding when the timer stops.
j. Momentarily depress reset switch S8 and reset time interval meter. All alarm indicator lights shall extinguish.
k. Momentarily set sensor simulator switch F1 to OPEN to simulate a duress alarm. Alarm indicator lights INST and DELAYED shall illuminate, the time interval meter shall begin running, and LED F on PWB A12 shall illuminate. The AA indicator light shall not light.
l. When the timer stops, it shall indicate 10 ±2 seconds. The sonalert shall stop sounding when the timer stops.
m. Momentarily depress reset switch S8. All alarm indicator lights shall extinguish.
n. Turn mode switch to SECURE and back to TEST/RESET. LED F shall extinguish. Reset time interval meter.
o. Set latch/nonlatch switch on status processor to NONLATCH.
p. Set timer function switch S3 to LATCH/NL, start trigger switch S4 to 2, and stop trigger switch S5 to 1.
q. Repeat steps d. through n. Responses shall be identical to those obtained in corresponding steps."

PAGE 50

Table XII:, delete in its entirety.

PAGE 56

1.c, delete and substitute:

"20 ohms, 50 watts, 5 percent resistor."

2.e, delete "15 ohm resister" and substitute "20 ohm resistor".

PAGE 57

3.h, line 1, delete "RS06" and substitute "RS03".

PAGE 59

2, perform following changes:

a. change steps "c." to "a.", "a." to "b.", and "b." to "c.".
b. delete step "h" in its entirety and replace with:
h. Turn on CU Power switch."

PAGE 60

3.b, delete in its entirety and substitute:

"b. Set Latch/Non-Latch switch to LATCH position."

3.h, line 4, delete "output 4" and substitute "output 3".

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PAGE 61

3.j, line 3, delete "Output 4" and substitute "Output 3".

PAGE 63

3.x, line 2, delete in its entirety.

3.y, line 4, delete "above 18.6 Vdc." and substitute "at no less than 18.3 Vdc."

3.z, lines 1 and 2, delete "sonalert quiets" and substitute "output 1 goes low".

3.z, line 4, delete "21.5 Vdc: and substitute "21.7 Vdc".

3.aa, delete in its entirety and substitute:

"aa. Place S15 in positon 1 and return power supply #2 to 24.5 + .5 Vdc. Replace fuse F2 and restore ac power to the CU."

3.bb, delete in its entirety and substitute:

"bb. At all times when tests are not being performed, the CU mode switch shall be in SECURE position, Latch/Non-latch switch in NONLATCH position, and oupputs 1 through 5 shall be continuously monitored.

PASS - Outputs 1,2 and 4 are low, outputs 3 and 5 high at all times.
FAIL - Nonconformance to PASS criteria."

PAGE 65

2.3e, delete "0.600 +0.35" and substitute "0.600 + .045"

PAGE 66

2.4.1e, delete in its entirety.

2.4.1f, delete in its entirety.

2.4.1g, line 1, delete first sentence.
line 2, delete "table I" and substitute "table XI".

2.4.2d, line 2, delete "column E" and substitute "column F".

PAGE 67

2.4.5d, delete "table I" and substitute "table XI".

2.4.6b, insert, after "power supply", "(PS2)".

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PAGE 68

2.4.9e, delete and substitute:

"e. The resistance across TB3-1 and TB3-2 shall be 200K + 20K ohms within 5 seconds after opening S1."

PAGE 72

2.e, line 4, delete "18.6 Vdc" and substitute "18.3 VDc".

2.e, line 5, delete "18.5 Vdc:" and substitute "18.3 Vdc".

2.f, line 3, delete "OFF" and substitute "ON".

2.f, line 4, delete "OFF" and substitute "ON".

2.g, line 1, delete "21.4 Vdc" and substitute "21.7 Vdc".

2.g, line 4, delete "21.4 Vdc" and substitute "21.7 Vdc".

2.g, line 5, delete "20.5 Vdc" and substitute "21.7 Vdc".

2.h, add the following:

"NOTE: Conditions of white light should be ignored during steps j thourgh n."

PAGE 88

Figure 1, delete and substitute new figure 1.

Figure 2, delete asnd substitute new figure 2.

PAGE 90

Figure 3, delete and substitute new figure 3.

PAGE 93

Figure 6, change resistor connected between TB2-1 an TB2-4 of the Control Unit from 15 ohm, 2 watt to 20 ohm, 50 watt.

PAGE 94

Figure 7, change the connection of R23 in the test fixture such that it is connccted between TB2-4 and TB2-6 instead of TB2-4 and TB2-5.

PAGE 96

Figures 9, step 3 of calibration procedure, delete "TB3-1" and substitute "TB2-3".

PAGE 98

Figure 11, delete and substitute new figure 11.

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PAGE 99

Figure 12, delete existing note and replace with the following:

"NOTE: Switch S7 on Test Set must be in position 1 for test of A1-A6 cards, and in position 2 for test of the A7 cards."

PAGE 101

Figure 14, On parts list: Change "IN751 ZENER OR EQ." to "NOT USED".

PAGE 105

Figure 18, Note: Change from "figure 21" to "figure 17".

PAGE 107

Figure 20, change "LAA" to "AA".

PAGE 108

Figure 21, delete and substitute new figure 21.

PAGE 109

Figure 22, delete list LED's and replace with:

"L#	TYPE
1,3,5,6,7,8	MONSANTO MV 5253 OR EQ
2,4	MONSANTO MV 5053 OR EQ
9	MONSANTO MV 5353 OR EQ

Figure 22, delete resistor table and replace with:

"R#	VALUE
1,2,4,11,17,23,24,25,30,37,40,41,42,43	47K ohms
3	TBD NOTE 1
5,10,14,16,20,22,26,27,28,35,38,44,45	10K ohms
6	TBD NOTE 2
7	TBD NOTE 2
8	TBD NOTE 3
9	TBD NOTE 3
12,48,50	4.7K ohms
13,15,18,19,21,29,36,39,46	330 ohms
31	270 ohms
32,33,34	IK ohm
47	15k ohms
49	TBD NOTE 4

ALL RESISTORS 1/4 WATT, +10% EXCEPT WHERE NOTED

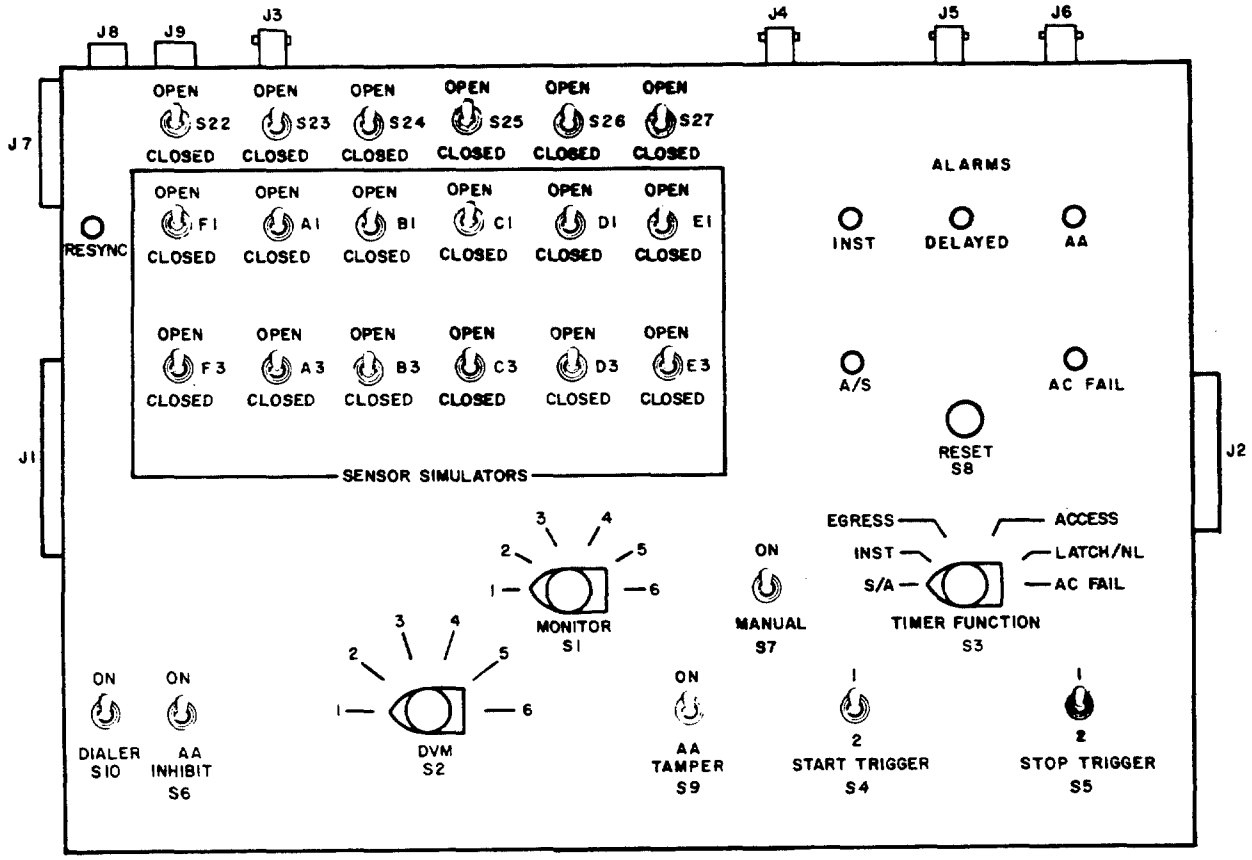
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NOTE:
SEE FIGURE 2 FOR TEST FIXTURE SCHEMATIC DIAGRAM.

FIGURE 1. Control unit test fixture.

X-2978 D

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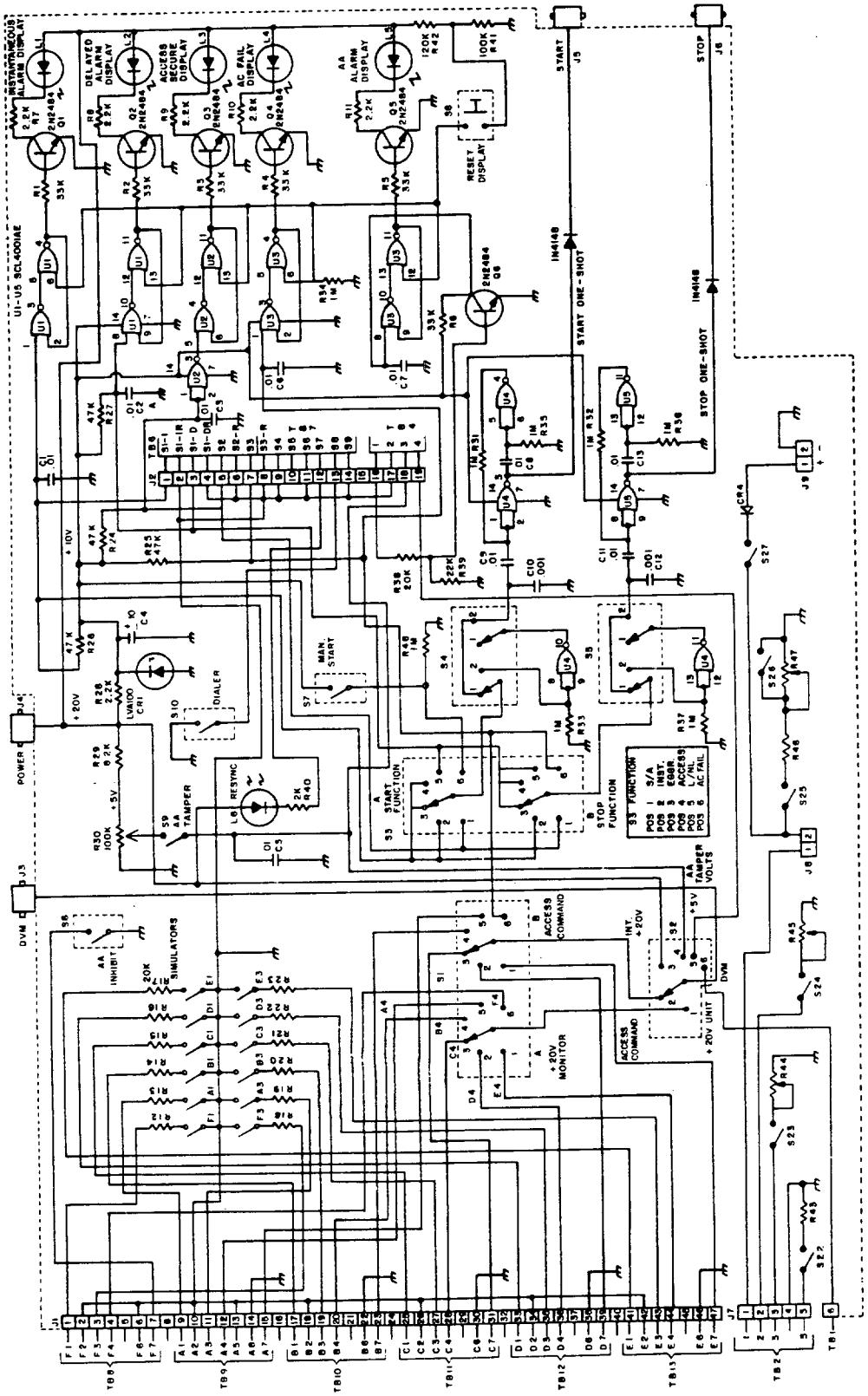


FIGURE 2. Control unit test fixture schematic.

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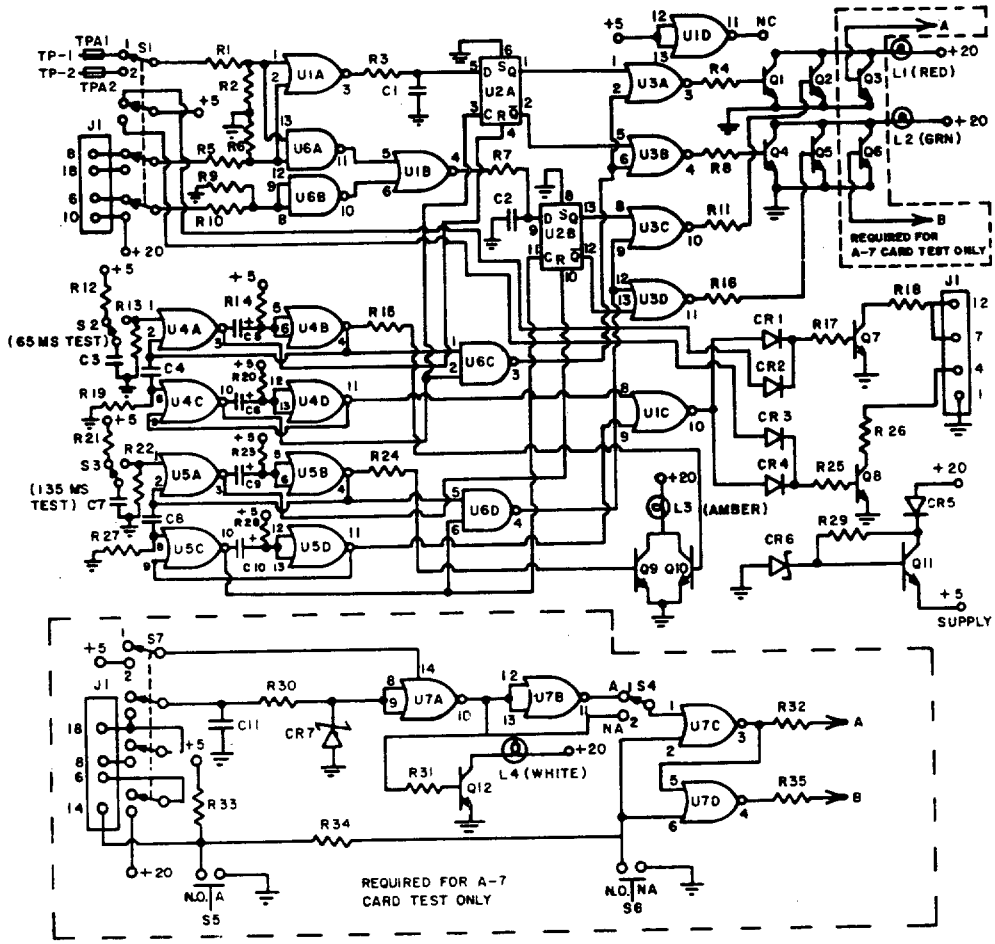
QTY	ITEM	DESCRIPTION	REF SYMBOLS
2	CAPACITOR	0.001 uF 50 V	C10, 12
11	CAPACITOR	0.01 uF 50V	C1-9, 11, 13
3	DIODE	IN4148 OR EQUIV	CR2, 3, 4
1	DIODE, ZENER (10.0V)	LVA100	CR1
6	DIODE, LIGHT EMITTING	MONSANTO MV5053 OR EQUIV	L1-6
5	INTEGRATED CIRCUIT	CD4001/MC14001 OR EQUIV	U1-5
1	RESISTOR	2,000 OHMS 5% 1/4W	R40
1	RESISTOR	3,000 OHMS 5% 1/4W	R49
6	RESISTOR	2,200 OHMS 5% 1/4W	R7-11, 28
1	RESISTOR	8,200 OHMS 5% 1/4W	R29
13	RESISTOR	20,000 OHMS 5% 1/4W	R12-23, 38
1	RESISTOR	22,000 OHMS 5% 1/4W	R39
6	RESISTOR	33,000 OHMS 5% 1/4W	R1-6
4	RESISTOR	47,000 OHMS 5% 1/4W	R24-27
1	RESISTOR	100,000 OHMS 5% 1/4W	R41
1	RESISTOR	120,000 OHMS 5% 1/4W	R42
8	RESISTOR	1 MEG OHMS 5% 1/4W	R31-37, 48
1	RESISTOR, VARIABLE	100,000 OHMS	R30
1	RESISTOR	100 OHMS 2% 1/2W	R43
1	RESISTOR, VARIABLE	50 OHMS 25W	R44
1	RESISTOR, VARIABLE	150 OHMS 12W	R45
1	RESISTOR	5 OHMS 5% 100W	R46
1	RESISTOR, VARIABLE	15 OHMS 100W	R47
2	SWITCH, DP6T ROTARY	CTS T206 OR EQUIV	S1, 3
2	SWITCH, DPDT TOGGLE	ALCO MTA206N OR EQUIV	S4, 5
1	SWITCH, SP6T ROTARY	CTS T205 OR EQUIV	S2
21	SWITCH, SPST TOGGLE	CUTLER HAMMER 7580K7 OR EQUIV	S6, 7, 9, 10, A1, B1, C1, D1, E1, F1, A3, B3, C3, D3, E3, F3, S22, S27
1	SWITCH, PUSHBUTTON	CUTLER HAMMER 8411K12 OR EQUIV	S8
11	NOT USED		S11-21
6	TRANSISTOR	2N2484A OR EQUIV	Q1-6

NOTES: 1. CALIBRATE RESISTORS R44, R45, AND R47 AS FOLLOWS (SWITCHES S22-S27 OPEN UNLESS OTHERWISE SPECIFIED):
   A. WITH AMMETER PLUGGED INTO J8, CLOSE S25 AND APPLY 27.5 VDC AT J7-1 (REFERENCE J7-4). ADJUST R47 FOR READING OF 1.70 + .02 AMPERES.
   B. OPEN S25, CLOSE S24 AND APPLY 27.5 VDC AT J7-2. ADJUST R45 FOR READING OF 250 ± 5 MA.
   C. OPEN S24, CLOSE S23 AND APPLY 20.0 VDC AT J7-3. ADJUST R44 FOR READING OF 600 ± 10 MA.
   2. PLACE R49 ACROSS J2-17 AND J2-18. ADJUST R30 FOR 4.5 ± 0.2 VDC ACROSS J2-17 (GROUND) AND J2-18 (POSITIVE). REMOVE R49 AFTER ADJUSTMENT.

FIGURE 3. Control unit test fixture parts list.

X-4288A

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U#	TYPE
1,3,4,5,7	#4001 QUAD 2 INPUT NOR GATE
2	#4013 DUAL "D" FLIP-FLOP W/SET/RESET
6	#4011 QUAD 2 INPUT NAND GATE

ALL CAPACITORS LOW LEAKAGE MYLAR, POLYCARBONATE, OR TANTALUM (±10% TOL.)

C#	VALUE	VOLTAGE
1,2	2 uF	12V
3,4,7,8	.01 uF	12V
5,9	43 uF	12V
6,10	2.2 uF	12V
11	.001 uF	25V

S#	TYPE
1,7	4PDT TOGGLE, CUTLER-HAMMER #SF4SCY191 OR EQ.
2,3	MOM. SPDT (BBM), GRAYHILL #46-101 OR EQ.
4	SPDT TOGGLE, CUTLER-HAMMER #8282K14 OR EQ.
5,6	MOM. SPST-N.O., GRAYHILL #39-1 OR EQ.

CR#	TYPE
1,2,3,4,5	IN456 OR EQ.
6	IN753 - 6.2V ZENER
7	IN751 - 5.1V ZENER

ALL RESISTORS 1/4W, ±10% TOL. (UNLESS NOTED).

R#	VALUE
1,5,10,30	15K
2,6,9	4.7K
3	NOTE 1
4,8,11,15,16,17,24,25, 31,32,35	10K
7	NOTE 2
12,13,19,21,22,27,34	47K
14,23	NOTE 3
18,26	20K
20	NOTE 4
28	NOTE 5
29	1K (1/2 WATT)
33	680Ω

J1 - CARD CONNECTOR, 18 PIN, ELCO 6007 OR EQ.

Q1 THRU Q12 - 2N2222A OR EQ.

L1 THRU L4 - #327, SYLV. #33309-0 OR EQ.

TPA 1, 2 - TIP PLUG, SMITH #492 OR EQ.

NOTE 1: R3 ≅ 13K, TBD SUCH THAT A 10.00 ± 0.05 MS AND GREATER OUTPUT AT U1A CAUSES U2A Q = 0 VDC AND U2A $\overline{Q}$ = +5VDC, WHEN U2A-C IS STROBED AT 100 KHz OR GREATER.
NOTE 2: R7 ≅ 21.5K, TBD SUCH THAT A 20.00 ± 0.10 MS AND GREATER OUTPUT AT U1B CAUSES U2B Q = +5VDC AND U2B $\overline{Q}$ = 0 VDC, WHEN U2B-C IS STROBED AT 100 KHz OR GREATER.
NOTE 3: R14, R23 ≅ 100K, SELECT FOR TIMER OUTPUT ≅ 2 SECONDS.
NOTE 4: R20 ≅ 70K, SELECT FOR 65± 2 MS PULSE DURATION AT U4D OUTPUT.
NOTE 5: R28 ≅ 94K, SELECT FOR 135±4 MS PULSE DURATION AT U5D OUTPUT.
NOTE 6: CIRCUIT TRACE LENGTH SHOULD BE MINIMUM PRACTICABLE TO AVOID CIRCUIT OSCILLATION.

FIGURE 11. A1-A6 and A7 cards-test fixture schematic. X-3912B

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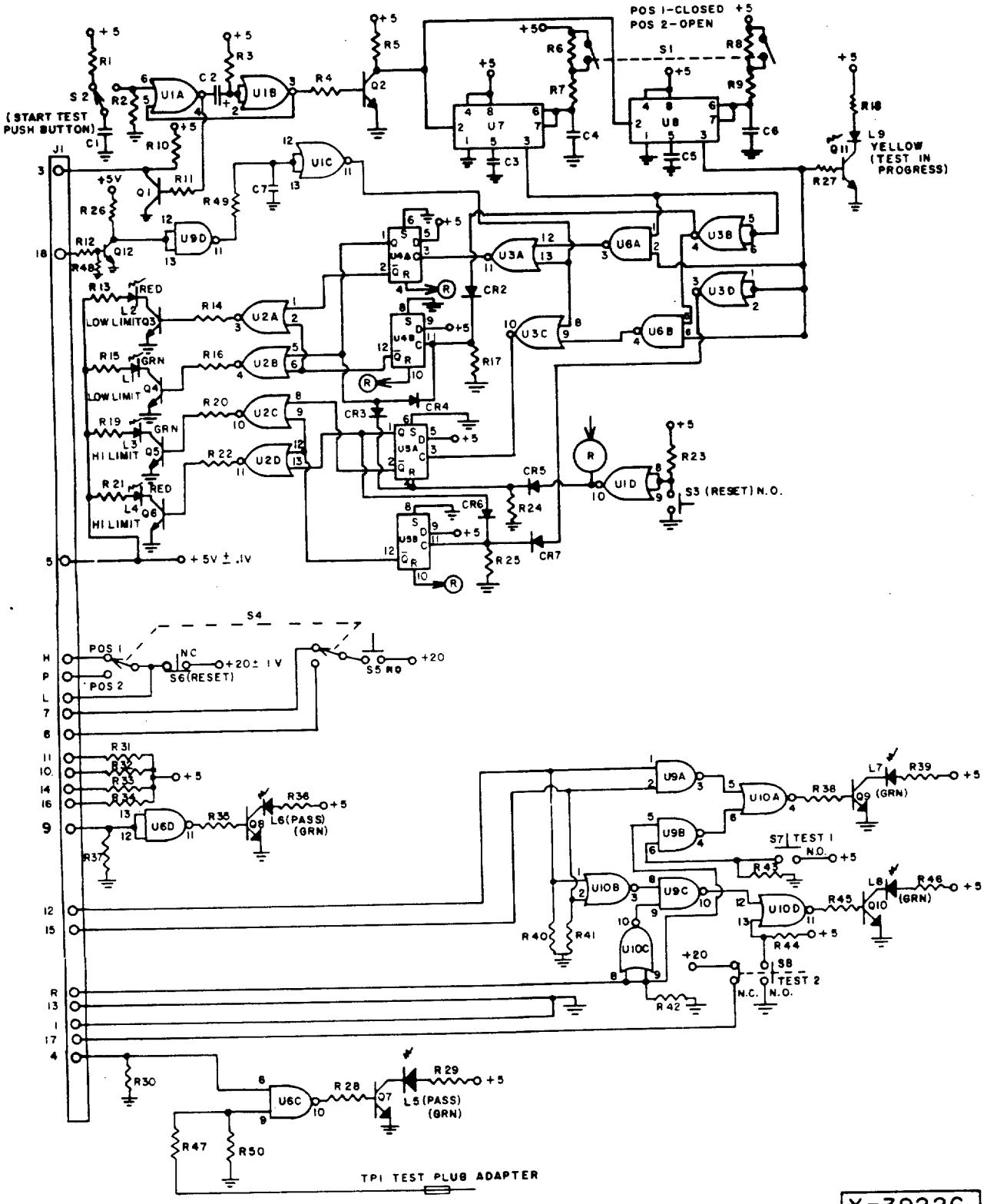


FIGURE 21. A10 card test fixture schematic.

X-3922C

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